

Simmon Yang

Richmond Hill, ON

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SUMMARY

Electrical Engineering Technology graduate (with Honours) with hands-on **electrical maintenance, repair**, and installation experience in **lighting** systems, **power distribution**, and **low-voltage** systems. Strong **troubleshooting** and **electrical code** compliance; coordinates effectively with licensed electricians and other **trades**. Safety-focused, with a full G driver's licence and own vehicle.

TECHNICAL SKILLS

- **Electrical & Hardware:** DC/AC circuits, **power distribution**, wiring, conduit, **lighting** fixtures and systems, panels and devices, testing & **troubleshooting** (continuity, voltage, insulation resistance)
- **Low-Voltage & Electronics:** Network Cabling (Cat5e/Cat6/Cat6A), Security Systems, PCB Design (KiCad), Microcontrollers (Arduino, Raspberry Pi), Soldering
- **Drawings & Tools:** Blueprint/Wiring Diagram Interpretation, AutoCAD Electrical, multimeters, hand and power tools
- **Standards & Safety:** **Electrical Code** Comprehension, WHMIS, Working at Heights, Lockout/Tagout

EDUCATION

Electrical Engineering Technology Diploma (with Honours)

Sep 2022 – Apr 2026

Centennial College

Scarborough, ON

- Relevant coursework: Electrical Theory, Drawing Interpretation & Installation, **Power Generation**, Monitoring Systems, Networking. GPA: 4.1

WORK EXPERIENCE

Electrician Assistant (As-Needed)

Sep 2024 – Apr 2026

Independent / On-call

Greater Toronto Area, ON

- Called in periodically to assist a licensed electrician on residential and commercial jobs during high-demand periods.
- Pulled, cut, routed, and terminated wiring/conduit for lighting, receptacles, and power distribution during rough-in and finishing phases.
- Installed switches, outlets, fixtures, and breakers to electrical code requirements, with correct grounding and secure device connections.
- Supported panel upgrades and device commissioning by verifying continuity, voltage, and insulation resistance before energization.

Low Voltage Technician

Jul 2025 – Sep 2025

OneNid Inc

Greater Toronto Area, ON

- Installed and terminated Cat5e/Cat6/Cat6A cabling for IP cameras, access points, and intercom systems to TIA/EIA standards.
- Integrated access control hardware (card readers, maglocks, motion sensors) with low-voltage wiring and control panels.
- Executed structured cabling from floor plans and wiring diagrams; labeled and organized runs for easier maintenance and troubleshooting.
- Validated terminations and diagnosed faults using cable testers, crimping tools, and multimeters before system handoff.

Electrical Technician Assistant

Jul 2021 – Aug 2021

Microgreen Solar Corp

Markham, ON

- Repaired and tested DC-AC inverter power packs for solar systems under multiple load conditions.
- Assembled and terminated custom power/signal cables, validating shielding, polarity, and connector integrity.
- Performed component-level PCB troubleshooting with multimeters, oscilloscopes, and soldering tools.
- Executed load and QA tests to verify voltage regulation, inverter performance, and safety compliance.

HVAC Assistant

May 2024 – Dec 2024

HVAC Contractor

Greater Toronto Area, ON

- Installed, maintained, and serviced residential/commercial HVAC systems, including wiring and electrical connections.
- Used wiring diagrams, schematics, and system layouts to troubleshoot installation issues and verify correct setup.

Residential Renovation Assistant

May 2023 – Aug 2023

Construction Inc

Greater Toronto Area, ON

- Installed electrical wiring for full-home renovations, including switches, outlets, and lighting fixture connections.
- Supported multidisciplinary renovation tasks (drywall, painting, demolition) to keep projects on schedule.

PROJECTS

Automated Circadian Lighting (Retrofit)

Sep 2025 – Dec 2025

Lighting Systems / Electrical

- Designed a retrofit LED lighting control system that adjusts color temperature and brightness throughout the day; developed a control profile transitioning from cool/high-intensity daytime lighting to warm/low-intensity evening lighting without replacing existing pot lights.
- Evaluated power consumption and lighting performance to validate energy savings and improved visual comfort.

4x4x4 RGB LED Cube

Dec 2021 – Jun 2023

Arduino / C++ / PCB Design

- Built a multiplexed RGB LED cube architecture that individually addresses 192 LEDs with 6 Arduino pins using shift registers and Bit Angle Modulation (BAM), avoiding per-LED driver ICs.
- Designed custom KiCad PCBs for the LED matrix and produced a 3D-printed enclosure for assembly.

CERTIFICATIONS

- WHMIS Certification (2026)
- Working at Heights Certification (2024)
- Full G Driver's Licence (Ontario), own vehicle